

ELKINS 21 ENE, WV, USA

WMO: 724370

Lat: 39.0130N

Lon: 79.4743W

Elev: 1033

StdP: 89.51

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03733

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -18.4	(c) -15.7	(d) -23.2	(e) 0.5	(f) -17.4	(g) -20.8	(h) 0.7	(i) -14.6	(j) 6.3	(k) -7.2	(l) 5.7	(m) -6.3	(n) 2.4	(o) N/A	(p) 0.448

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.6	(c) 25.8	(d) 19.2	(e) 24.9	(f) 18.5	(g) 23.8	(h) 18.0	(i) 20.7	(j) 24.3	(k) 20.0	(l) 23.3	(m) 19.4	(n) 22.3	(o) 1.9	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	16.3	22.5	18.9	15.6	21.8	18.3	15.0	21.1	64.8	24.5	62.3	23.1	59.8	22.4	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.0	4.5	3.9	-20.8	27.8	2.9	1.1	-22.9	28.6	-24.6	29.2	-26.3	29.8	-28.4	30.6
			-21.1	21.8	3.6	0.4	-23.7	22.1	-25.8	22.3	-27.8	22.5	-30.4	22.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-3.4	-2.4	2.2	8.5	13.4	16.7	18.4	17.9	15.2	9.6	3.7	-0.5
	DBStd	9.39	7.51	6.72	6.57	5.77	4.59	2.99	2.52	2.54	3.55	5.25	5.65	6.19
	HDD10.0	1744	419	349	255	94	23	0	0	0	4	73	198	329
	HDD18.3	3749	675	580	502	297	159	64	31	39	106	273	439	586
	CDD10.0	1133	2	2	12	49	128	202	261	244	159	60	9	3
	CDD18.3	93	0	0	0	2	5	16	33	25	10	2	0	0
	CDH23.3	309	0	0	0	9	22	49	104	78	42	6	0	0
	CDH26.7	10	0	0	0	0	0	2	6	1	1	0	0	0
Wind	WSAvg	1.8	2.3	2.4	2.2	2.1	1.7	1.4	1.2	1.1	1.2	1.6	1.8	2.1
Precipitation	PrecAvg	1132	88	81	105	96	115	109	119	98	87	75	67	92
	PrecMax	1662	166	173	186	197	240	182	223	196	223	129	142	159
	PrecMin	895	46	31	35	42	43	32	54	56	10	17	12	38
	PrecStd	173	28	33	38	38	47	40	45	36	53	37	30	32
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	15.9	19.8	24.7	25.4	26.3	27.6	26.7	26.2	24.1	19.4	15.4
		MCWB	11.2	11.9	11.5	13.0	17.0	19.5	21.4	20.4	18.9	16.8	11.0	12.0
	2%	DB	12.4	12.4	16.7	21.9	23.8	24.8	25.7	25.4	24.6	21.4	16.0	12.9
		MCWB	9.7	9.6	10.8	12.5	16.1	18.5	20.2	19.3	18.0	15.1	10.3	10.2
	5%	DB	10.2	10.2	14.2	19.6	22.2	23.5	24.5	24.1	23.0	19.5	13.8	10.8
		MCWB	7.4	7.6	9.3	11.8	15.9	17.8	19.5	18.5	17.6	14.5	9.0	8.7
	10%	DB	7.9	7.8	12.0	17.3	20.5	22.0	23.2	22.8	21.2	17.6	11.9	8.9
		MCWB	5.8	4.7	8.3	10.6	15.0	17.3	18.6	18.0	17.2	13.5	8.2	6.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	12.9	13.3	15.5	19.2	20.4	22.5	21.0	20.6	18.6	13.7	12.8
		MCDB	14.0	16.1	18.0	21.4	23.4	24.6	25.6	24.5	24.1	22.5	17.7	14.7
	2%	WB	10.2	11.3	11.8	13.9	17.9	19.4	21.2	20.2	19.6	17.5	11.9	11.3
		MCDB	12.1	13.2	15.0	18.8	21.9	23.0	25.0	23.4	23.0	20.2	14.6	13.0
	5%	WB	8.0	9.4	10.6	12.9	17.1	18.7	20.3	19.5	18.8	16.2	10.3	9.8
		MCDB	9.4	11.0	13.6	17.0	20.9	22.1	23.7	22.4	21.7	18.7	12.3	11.2
	10%	WB	5.4	6.8	9.3	11.7	16.3	18.1	19.7	18.8	17.9	14.5	8.6	8.3
		MCDB	6.7	8.7	11.9	15.6	19.7	21.0	22.6	21.5	20.4	16.9	10.9	9.8
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.4	8.9	10.3	9.2	8.6	8.6	9.0	9.8	9.3	8.7	7.5
		MCDBR	8.7	10.5	11.4	12.7	11.2	10.0	10.0	10.2	11.5	11.6	11.0	9.7
		MCWBR	7.4	7.9	6.9	6.2	5.0	5.2	4.8	5.2	5.4	5.8	6.8	7.1
	5% WB	MCDBR	9.1	10.4	9.8	9.7	9.0	8.4	8.6	8.2	8.8	8.8	9.1	8.8
		MCWBR	7.9	8.5	6.8	5.7	4.6	4.6	4.5	4.4	4.6	4.8	6.9	7.4
Clear-Sky Solar Irradiance	taub	0.270	0.286	0.310	0.351	0.384	0.419	0.433	0.443	0.358	0.319	0.287	0.270	
	taud	2.524	2.460	2.468	2.382	2.286	2.212	2.198	2.140	2.378	2.502	2.564	2.557	
	Ebn at Noon	922	946	952	925	895	861	844	825	886	895	894	897	
	Edh at Noon	74	91	101	117	132	142	143	148	109	87	71	67	
All-Sky Solar Radiation	RadAvg	1.81	2.62	3.63	4.71	5.22	5.78	5.60	5.11	4.26	3.12	2.21	1.60	
	RadStd	0.19	0.39	0.26	0.40	0.46	0.41	0.34	0.27	0.51	0.37	0.23	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (40 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	+61

Nomenclature: See separate page